Chemistry 3A – Fall 2025 Thursday Version**: Section 25811**

Name _____ ID _____

Mini-Midterm on Gases and Gas Law T**otal Points:** 28 Points (Extra Credit)

Instructions: Show all work for calculation problems to receive partial credit. Use proper significant figures where applicable. No notes or cards are allowed. You are allowed an approved calculator, Green & Yellow Sheets. You do not need scratch paper:put it on the sheet

1. (4 pts, 1 pt each) Match each mathematical formula or statement to its corresponding law by entering the correct letter in the blank.

Choices:

____ $V = k \times n$
____ $PV = nRT$
____ $\frac{P_1}{T_1} = \frac{P_2}{T_2}$
____ $P_1V_1 = P_2V_2$

(a) Boyle's Law
(b) Charles' Law
(c) Gay-Lussac's Law
(d) Avogadro's Law
(e) Ideal Gas Law

2. Standard Temperature and Pressure (STP) (3 pts)

(a) What is the numerical value and unit for Standard Pressure used in our course? (1 pt)

(b) What is standard temperature in Celsius (°C)? (1 pt)

(c) Convert that standard temperature into Kelvin (K). (1 pt)

3. Charles's Law Calculation (5 pts)

A flexible gas sample occupies a **volume** of $\mathbf{2.80\ L}$ at a **temperature** of $\mathbf{22^{\circ}C}$. The gas is heated at constant pressure until its new volume reaches $\mathbf{3.60\ L}$. What is the *new* **temperature** of the gas in **Celsius (**°C**)**? (Are intermediate conversions necessary?)

4. Kinetic Molecular Theory (3 pts)
According to the Kinetic Molecular Theory (KMT) for an ideal gas, describe what happens to the energy of gas particles during their collisions with each other and the container walls.

5. Ideal Gas Law Calculation (5 pts)
Calculate the **pressure** (in atmospheres) exerted by $\mathbf{0.150}$ **moles** of carbon dioxide gas (CO_2) confined in a $\mathbf{2.50}$ **L** container at a **temperature** of $\mathbf{310}$ **K**. (Use $R = 0.08205 \frac{\text{L·atm}}{\text{mol·K}}$)

6. Partial Pressures and Mole Fraction (8 pts)
A gaseous mixture in a sealed container consists of $\mathbf{0.300}$ moles of helium gas (He) and $\mathbf{0.200}$ moles of neon gas (Ne). The total pressure of the gas mixture is measured to be $\mathbf{1.50}$ atm.
(a) What is the total number of moles ($\boldsymbol{n}_{\text{tot}}$) of gas in the container? *(1 pt)*

(b) Determine the mole fraction ($\boldsymbol{\chi}$) of neon gas (Ne). *(2 pts)*

(c) Calculate the partial pressure ($\boldsymbol{P}_{\text{Ne}}$) of neon gas in the mixture. *(3 pts)*

(d) If some additional helium gas is injected into the container at constant temperature and volume, what happens to the mole fraction of neon (χ_{Ne})? Briefly explain why
(2 pts)